

**BUILD-
ING
METABOL-
ISM** RECIPES
FOR
FOOD
AND
RESOURCE
CYCLES

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ARETI MARKOPOULOU

**BUILDING METABOLISM;
RECIPES FOR FOOD & RESOURCE CYCLES**

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With the assistance of Sonia Sobrino Ralston

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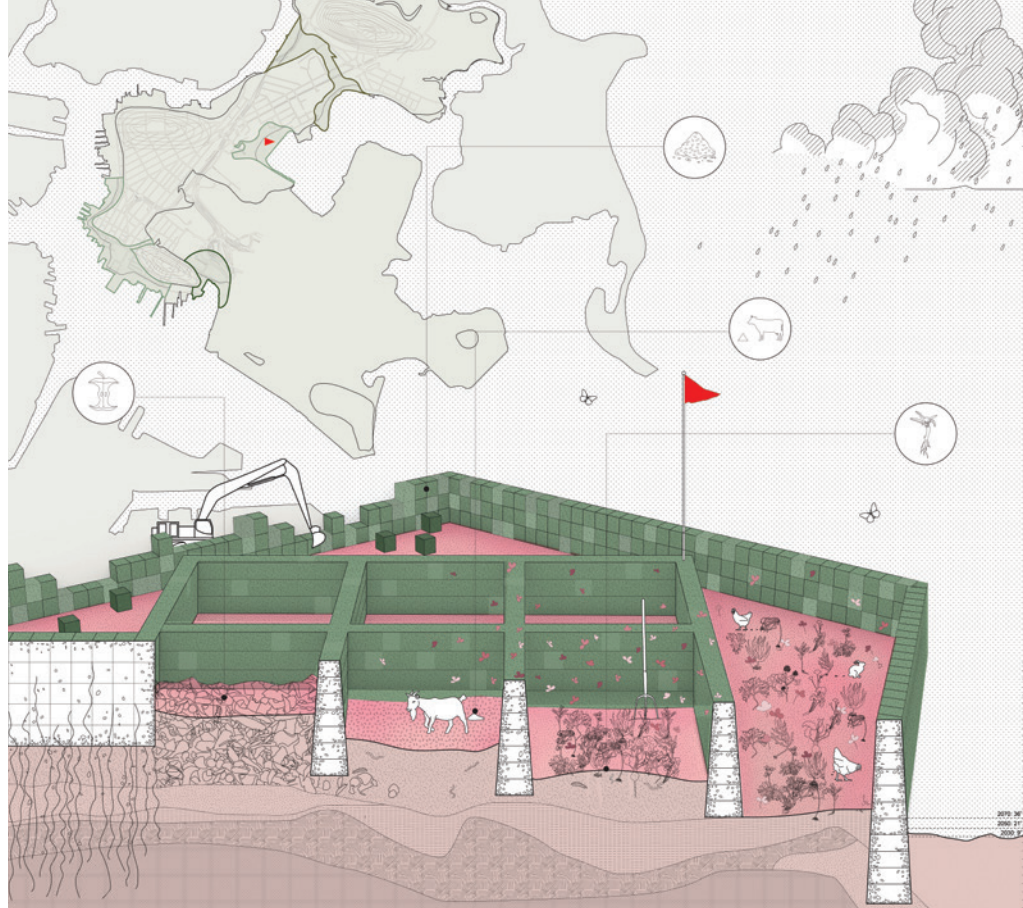
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SOLVING W/ORLDS

Design Earth
(Rania Ghosn and El Hadi Jazairy)

FIG. 01
Filling Islands drawing by Design Earth

“The destiny of a nation depends upon the manner in which they feed themselves,” wrote the French gastronome and philosopher Jean Anthelme Brillat-Savarin in his influential treatise on food, *The Physiology of Taste*. First published in 1826, this observation on how nations produce and consume food takes on further significance for the destiny not only of the nation, but also of the planet. How might we, in the midst of the sixth mass extinction, reflect on the destiny of the planet and the way some earthlings feed themselves? In their 2019 lecture titled “The Plantation and the Planet,” Donna Haraway and Anna Tsing proposed that the term “plantationocene” replaces the anthropocene as the fundamental transformation in deeply uneven “and patchy” ways of the natural-social worlds. In their work and conversations, the term “plantationocene,” comes to speak of the devastating transformation of different kinds of human-tended farms, pastures, and forests into enclosed and extractive plantations, relying on slave labor and other forms of exploited and migrant labor.

The destiny of a planet, one might argue, depends in part upon the manner in which its industrialists organize metabolic processes. The proclaimed bountiful abundance of the “plantationocene” has not only ecologically depleted the soils, but also increased hunger for the many with no access to sufficient, affordable and nutritious food. Best described as a paradox of the plenty, the resulting food insecurity is a systemic problem of access rather than scarcity. In the United States, for example, national per capita food supply is well above the recommended caloric intake and food waste is estimated at between 30 to 40% of the food supply. Meanwhile, more than 40 million people struggle to feed themselves. In cities, hunger is unevenly distributed along class, gender, racial, and ethnic lines, disproportionately affecting Black and Hispanic low-income households headed by women.

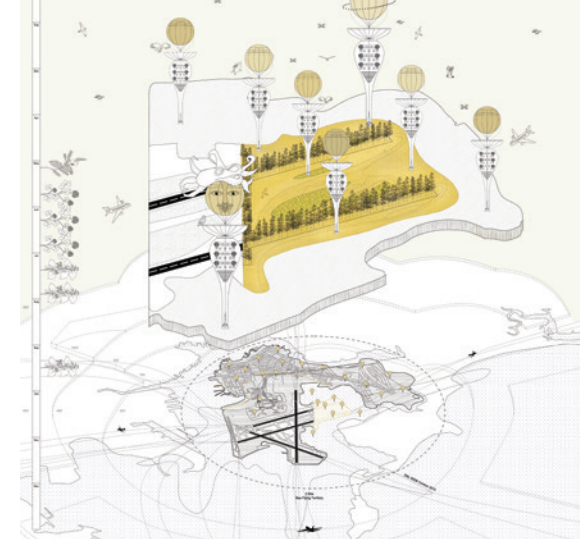


FIG. 03
Grounding Logan drawing by Design Earth

In a series of five drawings, the project “Sowing Worlds” proposes to reorganize forms of food metabolism in East Boston to sow sustenance and satiation and to replenish domains of water, soil, air that sustain life for all. The five drawings are titled “Mouth of the River,” “Filling Lands,” “Grounding Logan,” “Ghost Kitchens,” and “LifeStock.”

Mouth of the River

East Boston lies at the mouth of the Mystic River, where it flows into Boston Harbour. The name “MissiTuk,” meaning “great tidal river,” has been used to reference the temporal flooding of tidal flats, a geography that once supported a food web based on exchange between salt-water and freshwater. Land development, the expansion of chemical manufacturing, and the building of dams in the watershed have led to the spread of contaminants into the sediments and the sharp separation of saltwater and freshwater. “Mouth of the River” proposes to monitor the environmental stress on the Mystic River marsh and saltwater species across seven sites, with structures that sample the chemical contaminants, study the bioindicators, and grow salt-eating plants and species.

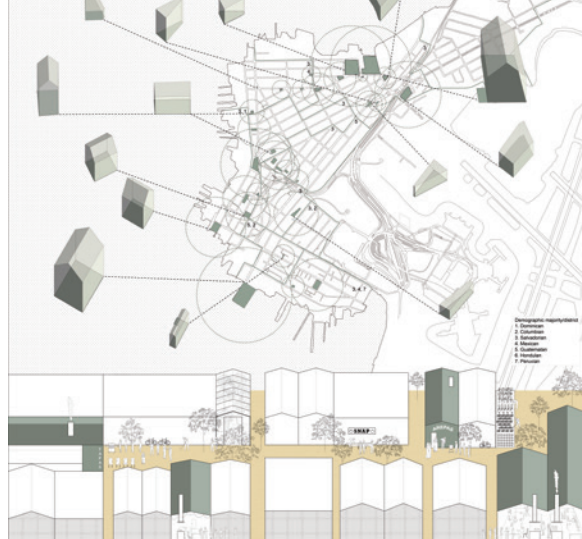


FIG. 02
Ghost Kitchens drawing by Design Earth

A metabolic approach to food-urbanism is a generative framework to examine the interrelations of social and ecological processes and ground them in specific contexts, all while potentially enabling designers to respond to metabolic processes across multiple spatial scales. In our project “Sowing Worlds,” East Boston is one such urban microcosm of food insecurity. The neighborhood is a historical critical entry point for immigrants to the US, with half of the population currently identifying as Hispanic or Latino. East Boston is also a spatial overlay and juxtaposition of numerous infrastructural sites—landfills, highways, tunnels, and Logan Airport—all of which present significant environmental externalities of noise, pollution, and now the dual threats of climate change with both increased flooding and green gentrification. The neighborhood is one of Boston’s two food deserts located in Suffolk County, where the average cost of a meal is 1.4 times the national average and the median household income is below the national average. East Boston is also a neighborhood with grassroots efforts and mobilizations to promote spaces for growing food and ensuring a livable climate for all.

Filling Lands

The 12 square kilometer area of today's East Boston was created by filling in the land between five inner harbor islands (Noddle's, Hog, Governors, Bird, and Apple), with a rich history of Indigenous life, then increasingly overlaid with practices of agriculture, industrial waste management, and military and detention centers. Rising sea levels and projections of future flood paths have reclaimed land.

"Filling Lands" recovers systemic organic waste generated by restaurants and food manufacturers from disposal and diverts it into compost for community garden sites. Such "land-fill" is framed by gabion walls and placed in such a way as to mitigate flood paths in East Boston.

Grounding Logan

Logan International Airport is the 16th busiest airport in the United States. It occupies approximately two thirds of East Boston, causing significant environmental impacts, such including the devaluation of nearby real properties, flight traffic noise, and air and soil pollution from jet fuel ultrafine particulates. Grassroot movements have been calling for local action on the climate impacts of aviation emissions. For example, Mothers Out Front East Boston called on the Massachusetts Port Authority to provide air filters, starting with public schools, and to plant trees—the outdoor air filters.

"Grounding Logan" proposes to restrict air traffic at the city's International Airport to limit air and soil pollution and to reclaim a runway as a forest for East Boston residents, including its birds and bees. The proposed honeybee farms would contribute to the pollination of nearby urban farms, provide an array of nutritious food products, and due to the high price of honey, ensure significant profit opportunities through beekeeping.

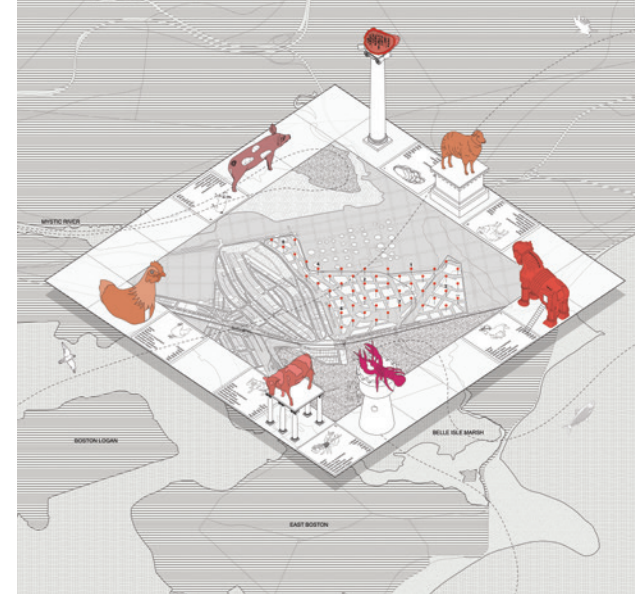


FIG. 04
Life Stock drawing by Design Earth

Ghost Kitchen

East Boston has been negatively affected by business closures and reduced health care support during the pandemic. The crisis response has also sparked a surge of grassroots initiatives and volunteer efforts to assess needs, deliver groceries, open kitchens, and prepare and distribute meals.

"Ghost Kitchens" transforms leftover odd city plots into pop-up community kitchens to expand options for food access, provide opportunities for small business, and incrementally increase sites for collective living and gathering, including collective childcare spaces to support the work of the parent-cook.

LifeStock

First planned in a grid-street style by the East Boston Company in 1833, East Boston has been significantly shaped by the logic of a developer-led urbanism, including relatively low land values that allow large parcels to be grabbed for redevelopment. More recently, the neighborhood has attracted developers because of its potential for strong commuter connections to downtown. The redevelopment of Suffolk Downs, the 161-acre former horse racing site on the border of East Boston, is one of the largest and most controversial real estate projects in Boston's history. In response, a coalition of community groups has outlined legal and equity concerns with the proposed redevelopment and its exclusionary processes.

“LifeStock” hacks the developer-driven building stock of East Boston through a series of Trojan-like farm animals on the site of a former racetrack. Each animal figure is funded through a standard format percent-for-art ordinance. In this allegorical tableau, the group of anthropomorphic farm animals rebel against their human developer.

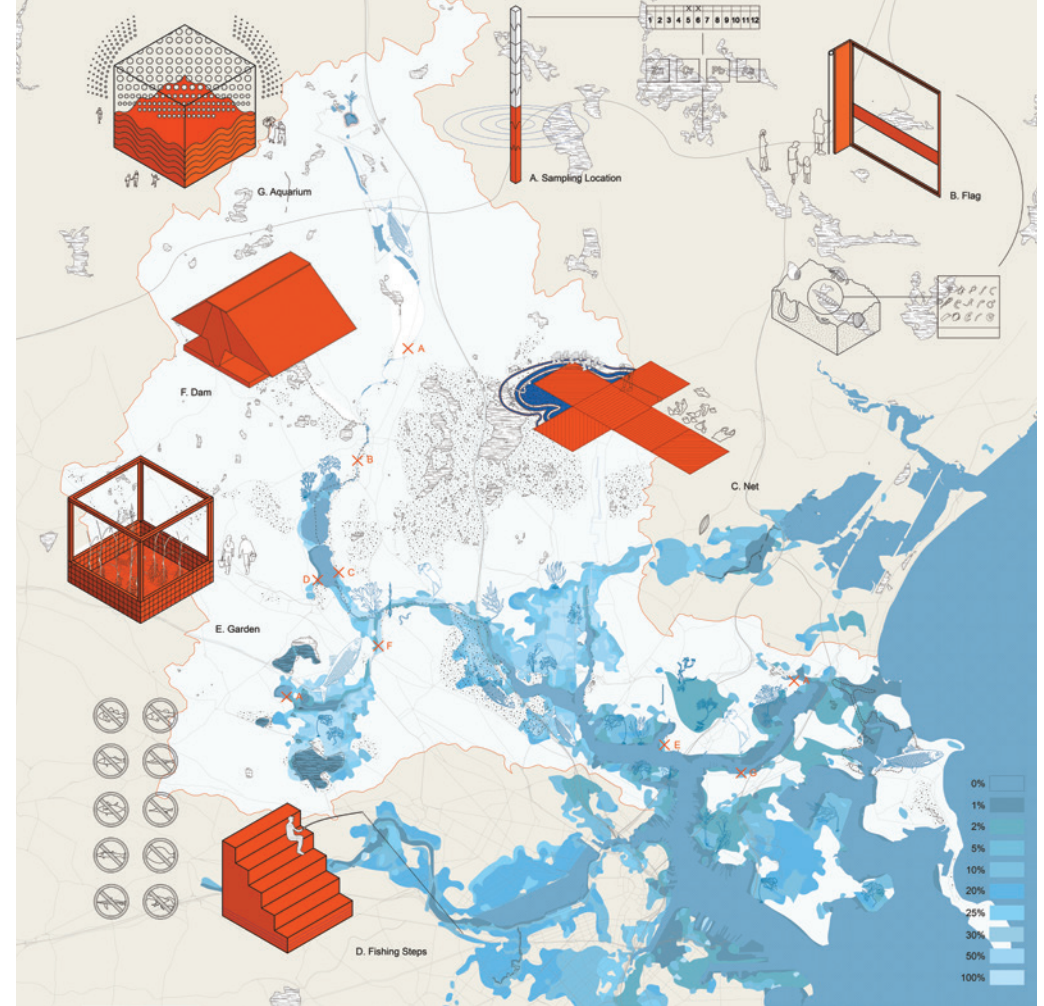


FIG. 05
Mouth of the River drawing by Design Earth

Project Credits

The Design Earth project team consisted of Rania Ghosn and El Hadi Jazairy, Melissa Gutierrez Soto, Eakapob Huangthanapan, Monica Hutton, Emma Jurczynski, and Anhong Li.

Images

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